

Practical 9:

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
GNU nano 8.7
int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3], n=12, frames=3;
    int i,j,k,hit=0,fault=0,pos,far;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            for(j=0;j<frames;j++) {
                if(frame[j] == -1) {
                    frame[j] = pages[i];
                    found = 1;
                    break;
                }
            }

            if(!found) {
                far = -1;
                pos = -1;

                for(j=0;j<frames;j++) {
                    int next = -1;

                    for(k=i+1;k<n;k++) {
                        if(frame[j] == pages[k]) {
                            next = k;
                            break;
                        }
                    }

                    if(next == -1) {
                        pos = j;
                        break;
                    }

                    if(next > far) {
                        far = next;
                        pos = j;
                    }
                }

                frame[pos] = pages[i];
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n", (float)hit/n*100, (float)fault/n*100);

    return 0;
}
```

OUTPUT:-

```
lenovo@LAPTOP-6627EU0U MSYS ~  
$ nano luu.c  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$ ls  
bankers.c  bankers.exe  dining.c  dining.exe  lu.c  lu.exe  luu.c  process  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$ gcc luu.c -o lu -pthread  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$ gcc luu.c -o luu -pthread  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$ ./luu  
Hits = 5  
Faults = 7  
Hit Rate = 41.67%  
Fault Rate = 58.33%  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$
```